Computer Networks - Chapter 3: Transport and Routing

Section 3.1: TCP (Transmission Control Protocol). TCP is a connection-oriented protocol, meaning a connection must be established between sender and receiver before data transfer begins, using a three-way handshake (SYN, SYN-ACK, ACK). TCP guarantees reliable delivery: it uses sequence numbers, acknowledgments, and retransmission of lost packets to ensure data arrives completely and in the correct order. Because of these guarantees, TCP is slower than UDP and has more overhead, making it suitable for applications like web browsing, email, and file transfer where accuracy matters more than speed.

Section 3.2: UDP (User Datagram Protocol). UDP is a connectionless protocol - there is no handshake and no guarantee that packets arrive, arrive in order, or arrive at all. This makes UDP much faster and lower overhead than TCP, since it skips acknowledgments and retransmissions entirely. UDP is commonly used for applications where speed matters more than perfect reliability, such as video streaming, online gaming, and voice calls, where a dropped packet is less harmful than a delayed one.

Section 3.3: Distance Vector Routing. In distance vector routing, each router maintains a table of distances (usually hop count) to every other router in the network, and periodically shares this table with its directly connected neighbors. Routers update their own tables based on what neighbors report, gradually converging on shortest paths. A well-known example is the Routing Information Protocol (RIP). A key weakness of distance vector routing is the 'count to infinity' problem, where routing loops can cause slow convergence after a link failure.

Section 3.4: Link State Routing. In link state routing, each router builds a complete map of the entire network topology by flooding information about its direct links to all other routers, not just its neighbors. Every router then independently runs a shortest-path algorithm (typically Dijkstra's algorithm) on this full map to determine the best route to each destination. Open Shortest Path First (OSPF) is a widely used link state protocol. Link state routing converges faster and avoids the routing loop problems of distance vector routing, but requires more memory and processing power since every router stores the full network topology.

Section 3.5: Comparing Distance Vector and Link State Routing. The core difference is what each router knows: distance vector routers only know distances reported by their neighbors, while link state routers know the complete topology of the network. This gives link state routing faster convergence and better loop avoidance, at the cost of higher memory and computational requirements. Distance vector routing is simpler to implement and requires less processing power, making it suitable for smaller networks, while link state routing scales better for larger, more complex networks.